ZIFENG ZHOU

(520) 339-3450 | zifengzhou01@gmail.com

EDUCATION

Rice University **Houston, TX**
Master of Computer Science *Aug 2026 - Present*

University of Arizona **Tucson, AZ**
B.S. in Computer Science, cum laude *Sep 2023 - May 2026*

- **Honors:** Dean's List (2x); Dean's List with Distinction
- **Relevant Coursework:** Python Programming Design (A), Discrete Mathematics (A), Algorithm (A), Parallelism and Distribution (A)

RESEARCH EXPERIENCE

Institute of Automation, Chinese Academy of Sciences (CASIA) **Remote**
Research Assistant, LLM Safety & RLHF *Nov 2025 - Present*

- **LLM alignment vs. robustness:** Investigated the structural gap between safety alignment and adversarial robustness; built a reproducible multi-stage pipeline covering benign alignment evaluation, prefix-injection attacks, adaptive attacks, JSONL/CSV processing, robustness-curve analysis, and cross-model comparison across open-source LLMs.
- **Robustness metrics and findings:** Designed and analyzed Prefix-k ASR curves, Robustness AUC, Adaptivity Gap, and alignment-robustness separation maps; experiments showed that stronger benign alignment does not necessarily imply stronger adversarial robustness, including degradation and ranking inversion under out-of-distribution attacks.
- **RLHF analysis:** Reviewed research on safety alignment, reward modeling, and multi-agent collaboration, and compared reinforcement-learning and alignment techniques including Q-value estimation and KL regularization to support experimental design and analysis.
- **Research manuscript:** Preparing a first-author manuscript for ICLR 2027 focused on trustworthy LLM evaluation and adversarial robustness.

University of Arizona **Tucson, AZ**
Research Assistant, Language Agent Tree Search *Mar 2025 - Sep 2025*

- **Evaluation pipeline:** Built data-processing and model-evaluation modules with Python, PyTorch, Hugging Face Transformers, and Pandas; developed test scripts for accuracy, precision, and error-distribution analysis.
- **Experimentation and debugging:** Deployed LATS experiments in Anaconda/Jupyter Notebook, used checkpoint logging and iterative debugging to resolve data-format and variable-passing issues, and visualized results with Matplotlib and Seaborn.
- **Results:** Ran LATS and baseline experiments on AIME and HotPotQA; observed **20%+ accuracy improvement** on discrete mathematics and logical paradox tasks and stronger performance on HotPotQA multi-hop reasoning.

PROFESSIONAL EXPERIENCE

Microsoft **Remote**
Artificial Intelligence Intern *Nov 2025 - Jan 2026*

- **Engineering environment:** Configured reproducible Windows/Linux environments using Miniconda, NVM, Git, SSH, GitHub Codespaces, and cloud GPU resources to support machine-learning experimentation, debugging, and collaborative development.
- **Machine learning:** Built and evaluated **XGBoost, TabNet, and SVM** models for enterprise valuation and loan-risk analysis, iterating through testing and validation to assess model performance and stability.
- **Generative AI and reproducibility:** Explored **Stable Diffusion and LoRA** pipelines for text-to-image and text-to-video generation; organized code, models, technical documentation, and structured experiment records on GitHub.

Global Impact Assessment (GIA) **Remote**
Artificial Intelligence Intern *Sep 2025 - Nov 2025*

- **Computer vision pipeline:** Preprocessed and visualized post-disaster satellite imagery with Python, OpenCV, and PyTorch using the xBD dataset; contributed to training, validation, and evaluation of **U-Net and RetinaNet** models.
- **Change detection:** Implemented pre/post-disaster change detection using pixel difference and SSIM, addressing preprocessing and inconsistent-result issues and proposing improvement approaches for complex-terrain scenes.
- **Visualization and reporting:** Integrated CNN predictions with geographic data to estimate affected populations, built Folium-based interactive damage maps, generated LLM-assisted disaster summaries, and wrote scripts for automated HTML comparison reports.

PROJECTS & ADDITIONAL EXPERIENCE

University of Arizona
*Research Assistant, **AI Medical Consultation Application*** *Jun 2025 - Aug 2025*

- **Full-stack development:** Developed React and Material UI components including dashboard navigation, report-printing controls, and ProtectedRoute logic for role- and permission-based access; integrated RESTful APIs for secure login, session tracking, and user-specific data loading.
- **Testing and delivery:** Used Postman, GitHub Issues, and PR reviews for API testing, debugging, task tracking, and collaboration; helped deliver a test version supporting role-based module access and viewing/exporting virtual-patient chat records.

University of Arizona **Tucson, AZ**
Undergraduate Teaching Assistant *Aug 2025 - Dec 2025*

- **Teaching support:** Supported **100+ undergraduate students** through lab and review sessions, tutoring, regular Q&A, and explanations of Python programming concepts and problem-solving approaches.
- **Course operations:** Graded programming assignments and exams and assisted with exam proctoring under faculty supervision while maintaining academic integrity.

SKILLS

Languages: Python, C++, C, C#, JavaScript, Java
ML/AI: PyTorch, Hugging Face Transformers, XGBoost, TabNet, SVM, U-Net, RetinaNet, Stable Diffusion, LoRA
Data / Development / Tools: Pandas, NumPy, OpenCV, Matplotlib, Seaborn, SQL, React, Material UI, Flask, Git, Linux, SSH, Jupyter Notebook, Postman
Spoken Languages: English (fluent), Mandarin (native), Cantonese (native)